

(A) 3D Multi-Modal Volumetric Input

(B) 3D Tokenization & Dual Masking

(C) ViT Context & Predictor

(D) Latent Collapse Prevention Paradigms

Raw BraTS 3D MRI

4 Channels: T1, T1c, T2, FLAIR
182 x 218 x 182 @ 1.0mm³

Canonical 3D Preprocessing

Non-Zero Brain BBox + Trilinear Resamp
Canonical Grid: 4 x 128 x 128 x 128

Parenchyma Normalization

Z-Score on Non-Zero Brain Voxels
Random Modality Dropout (p=0.25)

JEPAMaskingTransform3D

4 Target Cuboids (3x3x3, N_tgt ≈ 108)
3D BFS Context (N_ctx = 192, 37.5%)

PatchEmbed3D

512 Patches (8 x 8 x 8)
+ 3D Sinusoidal Pos Embed

ViT Context Encoder

Processes ONLY visible 192 tokens
85.9% Self-Attention FLOP Reduction

3D JEPA Predictor

4 Transformer Blocks (D=192)
Context Tokens + Target Pos Queries

3D I-JEPA (Assran et al., 2023)

Dual-Encoder + EMA Teacher ($m \in [0.996, 1.0]$)
Smooth L1 on LayerNormed Targets

3D SigReg JEPA (Balestrieri & LeCun, 2025)

Teacher-Free Single-Encoder + Projector (384 -> 1024 -> 128)
Epps-Pulley Gaussianity Test on M=256 1D Rays (Scale N)

3D VisReg JEPA (Wu et al., 2026)

Teacher-Free Single-Encoder + Projector
Decoupled Center + Scale $((1-\sigma)^2) + 1D$ SWD Shape (Stop-Grad σ)

(E) 3D Hierarchical Multi-Scale FPN Decoder

Hierarchical 3D Multi-Scale FPN Decoder

Extracts intermediate ViT representations from L2, L4, L6, L8 (Edge, Texture, Regional, Semantic Context)
Lateral 3D Skips + Progressive ConvTranspose3d + GroupNorm -> Predicted Whole Tumor Logits [1, 128, 128, 128]

(F) 3D Supervised Baselines

3D Residual UNet & 3D nnU-Net

MONAI 3D Residual UNet (InstanceNorm)
MONAI 3D DynUNet with Deep Supervision (Exponential Decay Weights $w_s = 2^{-s} / \sum 2^{-j}$)